Forestry: An International Journal of Forest Research, 2026, **99**, 1–17
https://doi.org/10.1093/forestry/cpaf067
Advance access publication date 6 November 2025
Original Article

National remote sensing-derived aboveground biomass yield curves for Canada

Piotr Tompalski*, Txomin Hermosilla, Sharad K. Baral, Michael A. Wulder, Joanne C. White

Canadian Forest Service (Pacific Forestry Centre), Natural Resources Canada, 506 West Burnside Road, Victoria, BC V8Z 1M5, Canada

*Corresponding author. Canadian Forest Service (Pacific Forestry Centre), Natural Resources Canada, 506 West Burnside Road, Victoria, BC V8Z 1M5, Canada. E-mail: piotr.tompalski@nrcan-rncan.gc.ca

Abstract

Accurate and current estimates of aboveground biomass (AGB) are essential outputs of forest inventories and are critical for carbon accounting. To obtain current estimates and provide projections into the future, growth simulators or yield curves are applied. Remotely sensed time series of AGB estimates across large areas provide a novel opportunity to generate representative AGB yield curves for a broader range of tree species, age classes, and regions, including noncommercial and unmanaged forests using a single, consistent data source and approach. In this study, remote sensing yield curves (RSYC) are demonstrated as a means to extend AGB modeling across Canada's forested ecosystems. To ensure national coverage while maintaining regional representativeness, yield curves were developed at a 150 × 150 km tile extent for 27 individual species, along with three additional multi-species models (generic, coniferous, and broadleaf). Evaluation against an independent set of plot data showed that among the multispecies models, the coniferous model achieved the lowest relative bias (2.06%) and root mean square error (RMSE) (41.03 t/ha), while species-specific models exhibited variable performance. Comparison to existing national species-specific yield curves indicated that the RSYC models generally achieved lower bias and RMSE than existing national models, with particularly notable improvements for key coniferous species such as black spruce and jack pine (e.g. RMSE% reductions from 55.89% to 34.69% for black spruce, and from 112.79% to 35.41% for jack pine), demonstrating enhanced accuracy and consistency for national-scale applications. Remotely sensed time series data enabled the development of a pool of species-specific and generic yield curves for Canada that are generated from a nationally consistent (and freely available) data source and approach—regardless of the management, ownership, or protection status of the forest.

Keywords: remote sensing; time series; Landsat; aboveground biomass; growth and yield

Introduction

Forests are an important component of the global carbon cycle, sequestering a significant proportion of anthropogenic carbon dioxide while also, under certain conditions, functioning as carbon sources (Pan et al. 2024). Accurate estimates of aboveground biomass (AGB) and related change are therefore critical for understanding and managing forest dynamics. This need is highlighted by international mandates to monitor and report forest carbon stocks and changes (Kurz et al. 2009, Birdsey and Pan 2015). Such information is essential for guiding climate mitigation strategies and ensuring compliance with national and global reporting obligations (Wulder et al. 2004, Kim et al. 2015). While well-established field plot networks often serve as the primary ground-truth source for carbon reporting, obtaining reliable estimates of AGB and its change over time across large areas remains challenging when relying on field plots alone, particularly in regions where plot coverage is spatially limited or temporally outdated. To address this, growth simulators and yield curves are widely used to extrapolate plot-level observations across broader spatial and temporal domains, and they play a fundamental role in both sustainable forest management and carbon monitoring. In forest management, they support harvest planning, logging, and thinning optimization, as well as assessments of ecosystem goods and services (Weiskittel et al. 2011, Shifley et al. 2017). In carbon monitoring, they form the basis for estimating biomass carbon stocks and fluxes, tracking emissions reductions, and informing climate mitigation policies (Kurz et al. 2002, Kurz et al. 2013). Examples include carbon budget models such as CBM-CFS3 (Kurz et al. 2009), EFISCEN (Schelhaas et al. 2007), CO2FIX (Masera et al. 2003), FullCAM (Richards and Evans 2004), or EFDM (Packalen et al. 2014), all of which can incorporate yield tables or empirical growth curves to simulate biomass and carbon dynamics.

Most existing approaches to projecting forest attributes such as AGB into the future rely on yield curves or growth simulators (growth and yield models). Yield curves represent the average expected stand volume or AGB development over time, whereas growth simulators can provide estimates for additional stand level attributes besides volume or AGB. However, growth simulators also require more complex input data and explicitly consider other factors such as competition and mortality (Weiskittel et al. 2011). Several growth and yield models have been developed in Canada, each differing in their methodologies, input data, and level of detail (Metsaranta et al. 2024). Most importantly, these models often focus on managed forests, overlooking the extensive areas of unmanaged forest stands accounting for a significant

Handling editor: Dr. Fabian Fassnacht
Received 3 June 2025; **revised** 18 August 2025; **accepted** 9 September 2025

portion of Canada's total forest area. In Canada, more southern, accessible, and higher productivity forests are typically under some form of management for timber or protection (Stinson et al. 2019). Over these managed forest areas, permanent and temporary sample plots are preferentially located and measured to develop growth and yield models. Due to a focus on timber production, these plots do not necessarily capture a representative range of forest types and structural conditions across Canada's forested landscape (Boisvenue and White 2019). Further, outside of the managed forests, plots are rare, with yield curves from productive forests often applied in these data-poor, unmanaged regions. Additionally, the application of these models over large areas remains complex, with ongoing efforts to harmonize them into a consistent modeling framework (Fortin and Lavoie 2022). As such, challenges remain, particularly with respect to the species represented and the geographic limitations of current yield curves and growth simulators.

Canada is a vast forest nation, with a diverse range of forest environments and conditions. Approximately 94% of its forest land is publicly owned, and forest management follows an extensive approach that contrasts with the more intensive approaches common in many European nations (Wulder et al. 2007, Fassnacht et al. 2023). Forest management responsibilities are vested with 12 unique jurisdictions in Canada, each of which maintains its own forest inventories and ground sampling programs (White et al. 2025). The development of yield curves or growth simulators at the national level is challenging because of the large ecological diversity that exists across the country. Development of national models would require the integration of data from multiple forest regions represented by different jurisdictions with different climatic conditions, tree species composition, and standards for field plot measurements and forest inventories (White et al. 2025). Incorporating such a broad range of conditions introduces considerable variability into any single modelling framework. While national models exist to estimate AGB (Lambert et al. 2005, Ung et al. 2008) or convert tree volume to AGB (Boudewyn et al. 2007), only a single national model currently provides estimates of AGB yield over time (Ung et al. 2009).

The national yield model developed by Ung et al. (2009) aimed to fill geographical gaps where regional models were lacking by generating projections of volume based on annual precipitation and degree-days. The model was built using a large dataset of permanent and temporary plots across Canada, focused on commercial species, with equations linking stand height, basal area, and volume to age and climate. Limitations to model development arose from the reliance on plots measured only once, which limited information on stand dynamics. As noted by the authors, the model is suitable only for short-term projections. Longer-term applications of the model without adjustments for stand density may lead to a biased estimate. Moreover, because the dataset used for model calibration emphasized commercially productive forests, model predictions may be biased outside the calibration range.

An alternative approach to generating yield curves was presented by Tompalski et al. (2024), who developed a methodology to derive remote sensing-based yield curves. By combining a time series of AGB estimates, forest age, and tree species data—each derived from Landsat data—the authors demonstrated that it is possible to produce locally representative yield curves for dominant boreal tree species. A filtering routine was implemented to identify and remove inconsistent or outlier AGB values, improving the input quality for the modeling process. Nonlinear mixed-effects models were then used to account for local and species-specific variability, producing robust yield curves over a 76.5 Mha study area located in Alberta, Canada. In total, once the study area was spatially partitioned, 127 yield curves representing eight species were generated.

While the study by Tompalski et al. (2024) confirmed the feasibility of deriving yield curves exclusively from time series of remotely sensed data, it also had several limitations. Although the approach was successfully demonstrated for eight species within a single study region, its applicability across Canada's full range of forest stand structures, species, and ecological conditions remained untested, and it was uncertain whether the method would generalize effectively at the national scale. The modeling approach relied on a model form where the same parameter determined both growth and decay, limiting the ability to capture early stand dynamics and senescence independently. Additionally, the model included a single random effect to account for pixel-level variation, without incorporating tile-level structure. As a result, a large and balanced sample across all age classes was required for each species within each tile to reliably fit yield curves. This limited the number of yield curves that could be developed, particularly in cases where available samples were concentrated in only part of the age range (e.g. older stands only). In addition, the evaluation approach was tailored to a regional field dataset and was not transferable to a national context.

Evaluating growth and yield models over large spatial extents remains challenging, particularly when reference data are sparse or unrepresentative of the full variability in forest structure, site conditions, and disturbance history (Vanclay and Skovsgaard 1997, Pretzsch 2009, Weiskittel et al. 2011). Previous studies have emphasized the importance of independent validation (Duncanson et al. 2021) and highlighted that small plot datasets often lack representation across the full range of stand ages, site qualities, and disturbance histories, complicating model evaluation efforts (Boisvenue and White 2019, Tompalski et al. 2024). Addressing these limitations requires a more comprehensive evaluation framework that accounts for the spatial and temporal mismatches between model outputs and validation data.

Considering the aforementioned context, the overall goal of this research was to develop and validate a suite of AGB yield curves with applicability across Canada's forested ecosystems. Specific objectives included: (i) refining and extending the methodology presented in Tompalski et al. (2024) to develop species-specific and multi-species yield curves at locally representative 150 km tiles; (ii) validating the models using independent field reference data; and (iii) comparing the resulting yield curves to national models published by Ung et al. (2009). The yield curves generated from this research offer a consistent framework for quantifying AGB across diverse ecological regions in Canada's forested ecosystems, supporting local, regional, and national applications.

Materials and methods

Study area

Forested ecosystems (~650 Mha) cover 65% of Canada's total land area and consist of forests, lakes, wetlands, and other wooded areas (Wulder et al. 2008). According to Canada's National Forest Inventory (2022), the treed area is ~370 Mha. Of this, ~62% is considered managed forest, with the remaining 38% classified as unmanaged (Natural Resources Canada, 2025) (Fig. 1). Forest composition is shaped by regional differences in climate, topography, soil conditions, disturbance history, and forest management practices, reflecting Canada's ecological diversity and wide range of forest conditions (Farrar 2017). Boreal forests in the north are

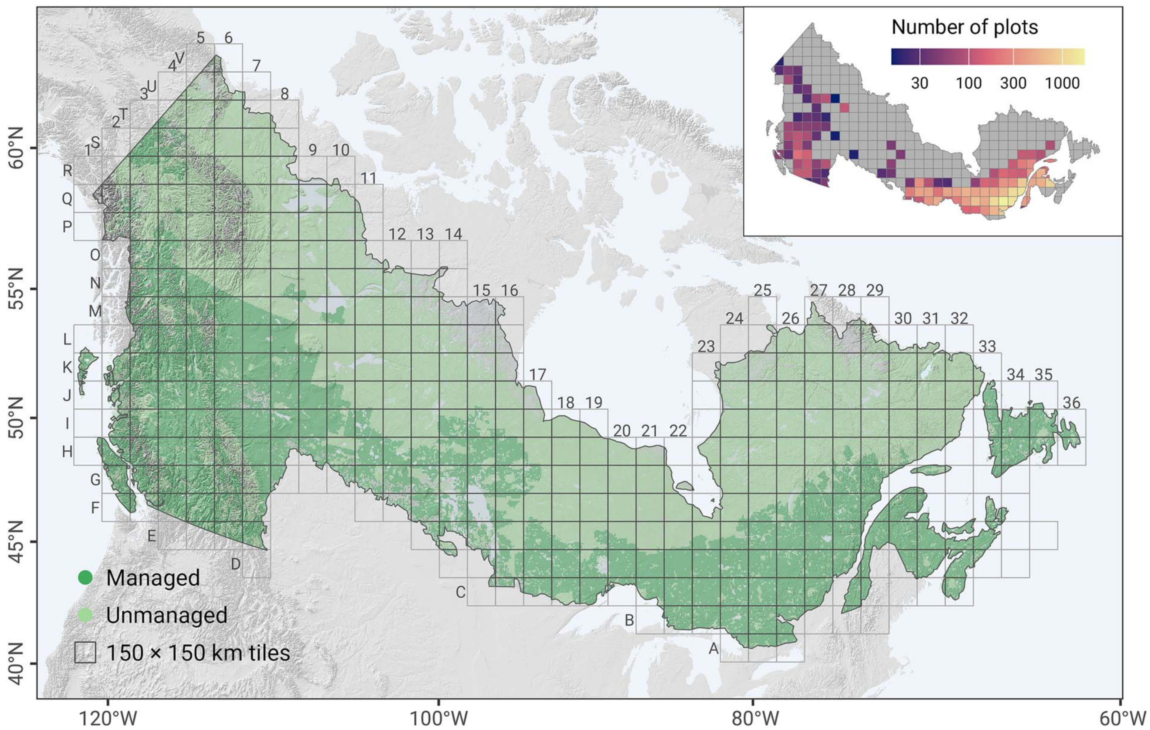

Figure 1. Canada's forested ecosystems map showing the 150 × 150 km tiling system, along with managed and unmanaged forest areas (after Stinson et al. (2019) and Wulder et al. (2020)). The inset (top right) shows the total number of field plots (MAGPlot) used during model evaluation.

primarily dominated by spruce (*Picea mariana, Picea glauca*) and pine (*Pinus banksiana*) species. West coast forests are characterized by hemlock (*Tsuga heterophylla*), cedar (*Thuja plicata*), Douglas-fir (*Pseudotsuga menziesii*), and subalpine fir (e.g. *Abies lasiocarpa*). In the temperate mixed forests of the central and eastern regions, sugar maple (*Acer saccharum*), yellow birch (*Betula alleghaniensis*), and trembling aspen (*Populus tremuloides*) are dominant. Disturbance regimes play a key role in shaping forest dynamics. While of high year-on-year variance, wildfire is the primary stand-replacing disturbance, burning an average of 1.61 Mha annually. Timber harvesting is common in the managed southern forests, with the affected area remaining relatively stable at an annual average of 0.64 Mha (White et al. 2017, Hermosilla et al. 2019).

Data

Remote sensing-derived data

Aboveground biomass and canopy height

We used a time series of annual AGB layers (AGB_{RS}) representing 39 years, from 1984 to 2022, where each layer contains 30 m spatial resolution pixel-level AGB estimates for a specific year. These layers were generated from remotely sensed data following the methodology described in Matasci et al. (2018a, 2018b). Their approach combined airborne laser scanning (ALS) data and field plot measurements to derive AGB (total aboveground biomass) estimates (Wulder et al. 2012), which were then spatially and temporally extended for all treed pixels (Hermosilla et al. 2022b) using annual Landsat surface reflectance, best-available-pixel (BAP) composites (White et al. 2014) and topographic data via nearest neighbor imputation. Independent validation indicated reasonable model performance for AGB with an R^2 of 0.70 and a root mean square error (RMSE) of 41.43 t/ha (65.8%) (Matasci et al. 2018b). Canopy height, as indicated by the 95th percentile of height (P95) from the ALS data, was derived via the same imputation approach as AGB, and its validation yielded an R^2 of 0.63 and an RMSE of 3.77 m (Matasci et al. 2018b). This estimated P95 was used to define post-strata during the validation phase of this study.

Tree species

Annual dominant tree species layers from 1984 to 2022 produced by Hermosilla et al. (2022a, 2024) representing 37 tree species across Canada's forested ecosystems were used as additional inputs. These maps were created using regional Random Forest models with training data from Canada's National Forest Inventory (NFI). A tile-based implementation of models, with tiles of 150 km × 150 km in size, ensured that predictions were locally adapted to species occurrence and environmental conditions. Predictor variables included annual Landsat surface-reflectance information, geographic, climatic, and topographic data, reflecting spatial and temporal dynamics in species composition. Post-classification processing was applied to the preliminary annual classification results in the form of a Hidden Markov Model in order to improve the temporal consistency of the tree species classification within the time series (Abercrombie and Friedl 2016). Validation with independent data indicated that the classification of dominant tree species achieved an overall accuracy of 86.1% (Hermosilla et al. 2024).

Forest disturbances

Forest disturbances from 1985 to 2021 were detected using the Composite2Change approach. This method analyzes spectral trends from annual Landsat surface reflectance BAP composites to identify the year of change (Hermosilla et al. 2015b) and temporal traits associated with the change, such as change magnitude and duration. A Random Forests classification model was used to attribute change events to a disturbance agent (fire, harvest, nonstand replacing), taking into account the spectral, temporal and geometric characteristics of the change events (Hermosilla et al. 2015a). Accuracy assessment showed that 90% of changes were correctly detected, with 89% identified in the correct year of occurrence and 98% identified within ±1 year. Disturbance attribution achieved an overall accuracy of 92% (Hermosilla et al. 2016).

Forest age

We applied forest age layers produced using the approach described in Maltman et al. (2023). This approach combines satellite-derived disturbance data and spectral information from the annual Landsat surface reflectance BAP composites. Maltman et al. (2023) integrated three different methods for age estimation: disturbance-based, recovery-based, and allometric-based. The disturbance-based method estimated forest age considering areas affected by stand-replacing disturbances since 1985 using the forest disturbance layers from Hermosilla et al. (2016). The recovery-based method estimated forest age for areas with evidence of spectral recovery following stand-replacing disturbances between 1965 and 1984 (White et al. 2017). The allometric-based method was applied to forested areas without disturbances or evidence of spectral recovery, using inverted site index equations and canopy height data from the forest structure layers (Matasci et al. 2018a, 2018b). Each of the three methods is associated with varying levels of uncertainty in the forest age estimation, with the lowest uncertainty in the disturbance-based estimates of age and the highest in the allometric estimates (i.e. uncertainty increases as forest age increases). A comparison with independent NFI data showed a median difference of five years for forest areas younger than 150 years, with regional variability reflecting disturbance regimes and productivity (Maltman et al. 2023).

Validation plots

The Multi-Agency Ground Plot (MAGPlot) database compiles forest ground plot data from across Canada to improve data accessibility for forest research. It integrates historical and current forest ground plots from Canada's NFI and permanent ground sample plot networks maintained by 12 provincial and territorial jurisdictions. The harmonized dataset spans from the 1920s to the 2020s and includes ~14 million tree measurements from over 52 000 plots. All data undergo quality control, standardization, and harmonization procedures to produce a unified, analysis-ready database (National Forest Inventory 2023). The MAGPlot data were used exclusively to provide independent reference information for RSYC model evaluation and were not used in any stage of model training.

Importantly, MAGPlot is a composite dataset derived from a wide range of monitoring programs developed independently by different agencies. These programs were established for various purposes, including long-term monitoring, operational forest management, silvicultural research, and inventory calibration, and followed different sampling designs. As a result, the MAGPlot database does not represent a single, unified sampling design and cannot be assumed to provide unbiased estimates at specific spatial units such as individual 150 × 150 km tiles.

Tree-level total AGB was estimated for the plots using allometric models based on species, diameter at breast height, and tree height measurements (Lambert et al. 2005, Ung et al. 2008), and subsequently aggregated to the plot level. Stand age for each plot was assigned using the Common Attribute Schema for Forest Resource Inventories (CASFRI) database, which integrates forest inventory polygons from different jurisdictions across the country (Canadian CASFRI Working Group 2025). CASFRI provides a harmonized data model that enables the use of forest inventory datasets collected under different standards by provincial, territorial, and federal agencies. The total number of plots per tile is shown in Fig. 1.

Existing yield curves

We used the "Nat1" model developed by Ung et al. (2009) to generate tile-level volume yield curves. The Nat1 model requires inputs such as species, growing degree days, and precipitation. Height and basal area are internally estimated by the model, and volume is subsequently derived using an allometric equation. For our inputs to Ung et al., we used 30-year climate normals (1971–2000) for growing degree days (>0°C) and total annual precipitation (Wang et al. 2016). Climate data were summarized at the tile level by calculating the median growing degree days and median precipitation for all tiles. The resulting yield curve was then converted to AGB using volume-to-biomass conversion models (Boudewyn et al. 2007). We refer to these yield curves as Ung2009 throughout the rest of the manuscript.

Yield curve development

We utilized time series data on AGB, stand age, and species composition to develop remote sensing-based AGB yield curves (RSYC). Building on the approach of Tompalski et al. (2024), we adapted the methodology for application at the national scale. While the core methodology of Tompalski et al. (2024) remained consistent, we introduced key modifications to the method to improve its applicability nationally. These included the development of models representing additional species groups, enhancements to filtering and sample selection procedures, refinements to the nonlinear mixed-effects modeling approach, and a more robust validation framework. An overview of the yield curve development workflow is presented in Fig. 2.

Two sets of RSYC models were developed: species-specific, and multi-species. Species-specific models ($RSYC_{SP}$) were created for each species that was well represented across a range of stand ages at the tile level. Multi-species models included forest-type models that were developed for coniferous ($RSYC_C$) and broadleaf ($RSYC_B$) species groups, as well as a generic model ($RSYC_G$) developed by combining all available samples across species. Because $RSYC_G$ models relied on pooled data rather than species-specific datasets, they were more robust to sample size limitations, converged more reliably during model fitting, and were used to provide initial parameter estimates for species-specific models. Additionally, $RSYC_G$ allowed for the characterization of AGB dynamics across mixed-species forests or in cases where species-specific information was unavailable or unnecessary.

Sample pool selection

A sample pool was established to ensure that only pixels with reliable pixel-level time series of AGB values were used for model development. Disturbed pixels were divided into pre- and post-disturbance time series and analyzed independently. The sample pool included pixels with at least 10 years of observations, that were not located at the edges of contiguous pixel groups belonging to the same tree species, and that maintained the same species classification for at least 50% of the time series.

To eliminate outliers, a maximum AGB value was determined for each processing tile based on all available MAGPlot data within

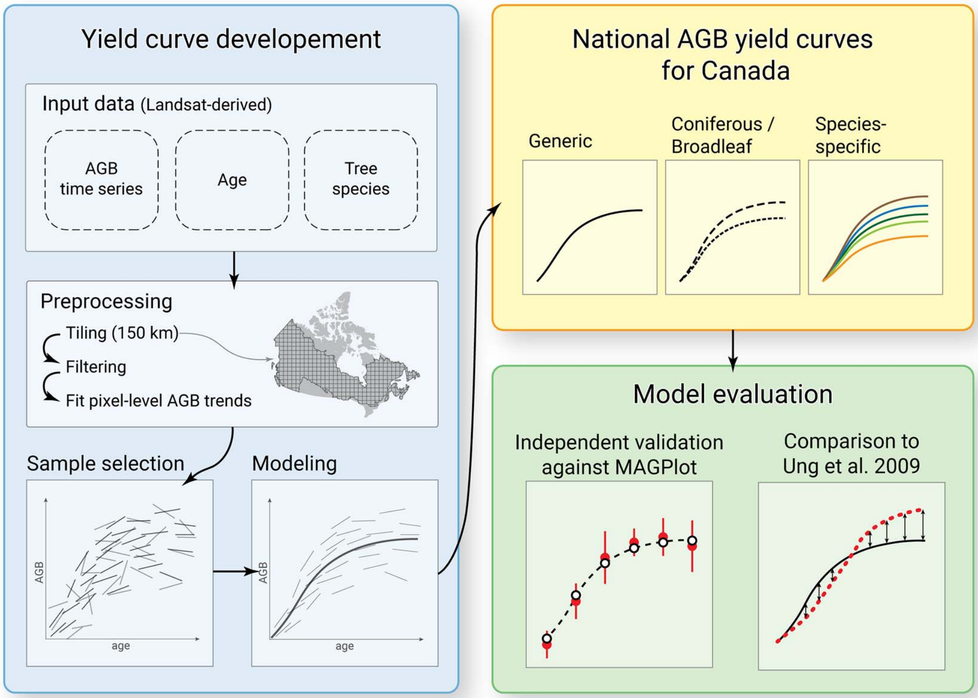

Figure 2. Workflow for the development of national remote sensing yield curves (RSYC).

the tile. For tiles without plot data, this value was interpolated using inverse distance weighting, based on the maximum observed AGB values from the three nearest neighboring tiles with available plot data. The final AGB threshold for outlier removal for each tile was calculated by rounding the interpolated or observed maximum AGB to the nearest 50 t/ha and adding 100 t/ha (e.g. if the maximum AGB observed at MAGPlot was 320 t/ha, the final threshold was 450 t/ha). Any pixels having outlier AGB values exceeding the tile-specific threshold were excluded from our analyses. Across all tiles, this filtering step affected 0.06% AGB values in 0.98% of all treed pixels. Pixel-level AGB trendlines for remaining pixels were then fitted and used as input to the RSYC model.

A balanced dataset for AGB modeling was selected using a stratified random sampling approach. Sampling was conducted separately for each combination of tile and species (hereafter referred to as tile-species stratum). Within each stratum, pixels were further categorized into 10-year age classes based on the initial age of their AGB time series. From each age class, 100 pixels were randomly chosen. To mitigate spatial autocorrelation, sampled pixels were required to be at least 500 m apart. In total, 1600 AGB trendlines (100 pixels × 16 age classes) formed a chronosequence of AGB for each species within each tile (Tompalski et al. 2024).

Model development and fitting

A nonlinear mixed-effects modeling framework was used to model the AGB as a function of age. A four-parameter Chapman–Richards derivative model form was used to allow for two inflection points in the developed curves. The model form was as follows (Equation 1).

$$y = \beta_1 e^{-\beta_4 age}\left(1 - e^{-\beta_2 age}\right)^{\beta_3} \tag{1}$$

where y is the AGB in t/ha^{-1}, age is pixel-level forest age in years, β_1, β_2, β_3, and β_4 are parameters to be estimated that set the asymptote, growth rate, shape, and decay, respectively.

We introduced a nested random effect structure to improve the model's representation of local conditions. The unique tile identifier was used as the first random effect, allowing the model to account for geographical variation across tiles. Nested within this, the unique pixel identifier was used as the second random effect to address the issue of repeated observations within the same pixel, which would otherwise violate the assumption of data independence.

Nonlinear mixed-effects models (Equation 2) were fitted for each species and species categories to get parameter estimates required for species specific and generic yield curves. The stochastic component of the model was defined by including the random effects on the β_1 parameter. We model the AGB observation y_{ijk} at year k, in the j-th individual pixel nested within the i-th tile, using the following nonlinear mixed-effects formulation (Equation 2):

$$y_{ijk} = \left(\beta_1 + b_i + b_{ij}\right)\ e^{-\beta_4 age_{ijk}}\left(1 - e^{-\beta_2 age_{ijk}}\right)^{\beta_3} + \epsilon_{ijk} \tag{2}$$

β_1–β_4 are considered fixed parameters; b_i and b_{ij} are random effect parameters specific to tile and pixels nested within a tile, respectively, and ϵ_{ijk} is the within group random error assumed to be independent of the random effects (i.e. we assume ϵ_{ijk} *i.i.d.* $\sim N\left(0, \sigma^2_{ijk}\right)$, $b_i \sim N\left(0, \sigma^2_i\right)$, and $b_{ij} \sim N\left(0, \sigma^2_{ij}\right)$). By incorporating pixel-level nesting within tiles and using age as a temporal covariate, the model captures both spatial and temporal variation in AGB. All models were fitted using the nlme (Pinheiro et al. 2023) package in R version 4.2.2 (R Core Team 2024). Model fit diagnostics (e.g. assumption of constant variance of residuals, normality of residuals and random effects) were visually

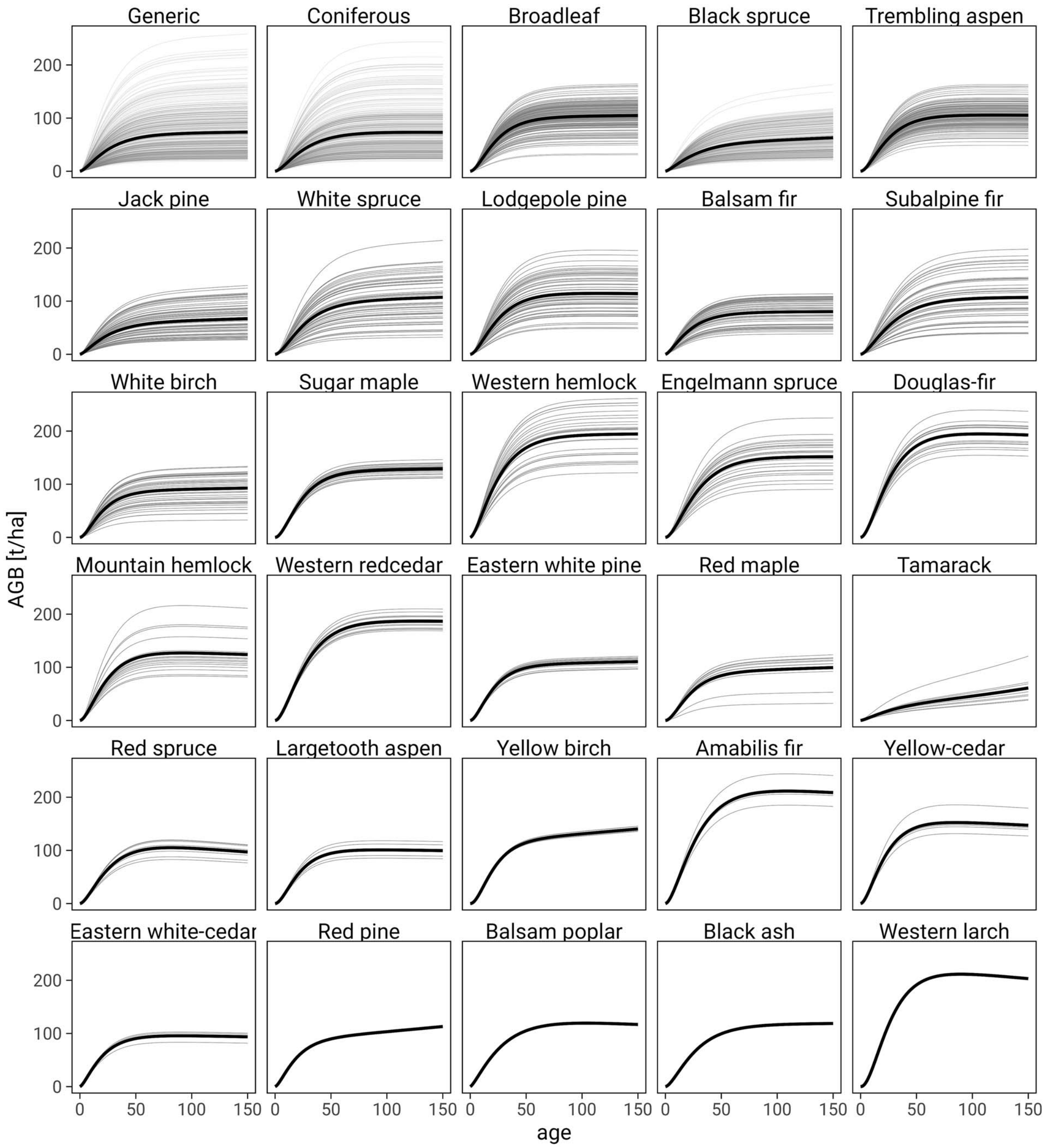

Figure 3. Developed AGB yield curves. Thin grey lines represent tile-specific yield curves with the random effect included, while the thick black lines show the models without the random effect. Note that in cases where a species occurs on only one or two tiles, the tile-specific yield curves (grey lines) may be covered by the black line and thus not visible (i.e. red pine, balsam poplar, black ash, and western larch).

assessed and no serious violation of the model assumptions was detected. Goodness of model fit was characterized by conditional and marginal R^2 (Nakagawa and Schielzeth 2013) and the RMSE.

Model evaluation

Model evaluation was performed by comparing the mean AGB estimates from the RSYC models to mean field-measured AGB, and involved stratifying AGB values into 20-year age classes. All model variants ($RSYC_{SP}$, $RSYC_G$, $RSYC_C$, $RSYC_B$) were assessed in tiles with a sufficient number of field plots (minimum three plots per age class, and at least four age classes). Models were evaluated at both the tile and strata level, where strata refer to the model variants representing either multi-species groupings (Generic, Coniferous, Broadleaf) or species-specific models (e.g. Black spruce, Trembling aspen, Jack pine). Tile-level evaluation allowed for the identification of spatial patterns in model behavior, while strata-level evaluation provided aggregate performance metrics for each model variant. The evaluation followed the recommendations of Vanclay and Skovsgaard (1997), assessing model performance not only empirically, but also in terms of biological realism and consistency with established knowledge of forest growth. None of the field plots used for model evaluation were used in the training of the remote sensing-based inputs (AGB, age, or species).

The MAGPlot data were used to evaluate tile-specific yield curves in the maximum number of tiles possible. Although these plots are established to be representative of specific forest conditions (e.g. species, density, stand age and site), they cannot be confirmed as probability samples representative of the corresponding population. As MAGPlot integrates data from various programs with differing objectives and sampling intensities, spatial coverage and representativeness vary across regions and jurisdictions. Hence, we expect sampling bias in the MAGPlot database as a whole.

A poststratification weighting approach (i.e. the raking ratio method) was used to minimize sampling bias by adjusting sample weights to a known population distribution of an auxiliary variable in each tile, with P95 serving as the auxiliary variable to define post-strata. P95 was selected because, as a direct measure of lidar return heights, it is a robust indicator of forest structure (Anderson et al. 2006), is correlated with AGB, but was not used in developing the RSYC yield curves, thereby allowing for exogenous poststratification in the weighting procedure (Magnussen et al. 2015). First, all plots and pixels within each tile were stratified by species into seven 20-year age classes, ranging from 10 to 150 years. Within each age class, samples were further stratified into three height classes ("short", "medium", and "tall") based on P95 values, using the 0.33 and 0.66 quantiles as class breaks. A weight (w_i) was then calculated for each height class i as follows (Equation 3):

$$w_i = \frac{RF_{px,i}}{RF_{gp,i}} \tag{3}$$

where $RF_{px,i}$ is the relative frequency of pixels in height class i (i.e. the number of pixels in height class i divided by the total number of pixels across all height classes within a given age class), and $RF_{gp,i}$ is the corresponding relative frequency of ground plots in height class. The adjusted weight (w_{ij}) for each ground plot j in height class i was obtained by multiplying the relative weight (population proportion divided by the identically distributed sample proportion which is assumed to be 1) specific to ground plot j by the height class weight (w_i).

The weighted mean AGB after adjusting the sample weights to the population distribution for a given age class was estimated using Equation (4) and the standard deviation of the weighted mean AGB approximated as Equation (5):

$$\overline{X}_w = \frac{\sum_{i=1}^{n}\sum_{j=1}^{k} w_{ij}X_{ij}}{\sum_{i=1}^{n}\sum_{j=1}^{k} w_{ij}} \tag{4}$$

$$SD_w = \sqrt{\frac{\sum_{i=1}^{n}\sum_{j=1}^{k} w_{ij}\left(X_{ij} - \overline{X}_w\right)^2}{\sum_{i=1}^{n}\sum_{j=1}^{k} w_{ij} - 1}} \tag{5}$$

where X_{ij} is the AGB ground plot j height class i.

To assess model performance, we computed the coefficient of determination (R^2), bias, and RMSE. Relative bias and RMSE were expressed as a percentage of the observed mean. Additionally, we derived age class-level confidence intervals based on plot data and evaluated the proportion (P) of RSYC mean values that fell within these confidence intervals. Finally, the coefficient of variation (CV), calculated as $CV = \sigma/\mu$, where σ is the standard deviation and μ is the mean, was computed for relative bias and relative RMSE to quantify the variability of tile-specific results.

The R^2 was calculated using the following formula:

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{N}\left(y_i - \overline{y}\right)^2} \tag{6}$$

where SSE is sum of square error, SST is total sum of squares, y is the observed AGB, $\hat{y}$ is the predicted AGB, and $\overline{y}$ is the mean of observed AGB.

Bias and RMSE were calculated as follows:

$$bias = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right) \tag{7}$$

$$bias\% = \frac{bias}{\overline{y}} * 100 \tag{8}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2} \tag{9}$$

$$RMSE\% = \frac{RMSE}{\overline{y}} * 100 \tag{10}$$

where N is the total number of individual strata, and y_i is the observed AGB, $\hat{y}_i$ is the predicted AGB.

To provide a descriptive indicator of agreement between model predictions and field data that accounts for uncertainty, we also calculated the proportion of RSYC means falling within the 95% confidence intervals of the field-based estimates. The proportion P was calculated as follows:

$$P = \frac{\sum_{i=1}^{N} 1\left(L_i \le \overline{AGB}_{RSYC,i} \le U_i\right)}{N} \tag{11}$$

where P is the proportion of RSYC means within plot-level confidence intervals, N is the total number of strata, $1(\cdot)$ is the indicator function, which equals 1 if the condition inside is true and 0 otherwise, L_i and U_i are the lower and upper bounds of the confidence interval for stratum i, $\overline{AGB}_{RSYC,i}$ is the AGB_{RSYC} mean value for stratum i.

Comparison to Ung et al. models

We validated the Ung et al. 2009 yield curves using the same validation routine as for the RSYC models. Specifically, AGB values for the 20-year age classes were compared with plot-level estimates at the strata level. In addition to bias, RMSE, and R^2, we also reported the proportion of Ung2009 estimates that fell within the 95% confidence intervals of the mean field-measured AGB for each age class, just as we did for the RSYC models. Finally, RSYC and Ung2009 yield curves were compared through visual inspection, allowing for further examination of model trajectories, AGB values, and their variability across tiles, age classes, and species.

Results

National aboveground biomass yield curves

A total of 1892 yield curves were developed (Table 1, Fig. 3). $RSYC_G$ yield curves were generated for 385 tiles, while $RSYC_C$ and $RSYC_B$ yield curves were developed for 355 and 184 tiles, respectively (Figs 4 and 5). Species-specific models were created for 27 species out of 37 species available in the Hermosilla et al. (2024) dataset, with substantial variation in the number of tiles represented for each species. Some species had models developed for only a single tile (black ash, western larch, balsam poplar) whereas black spruce had the greatest number of tiles with developed

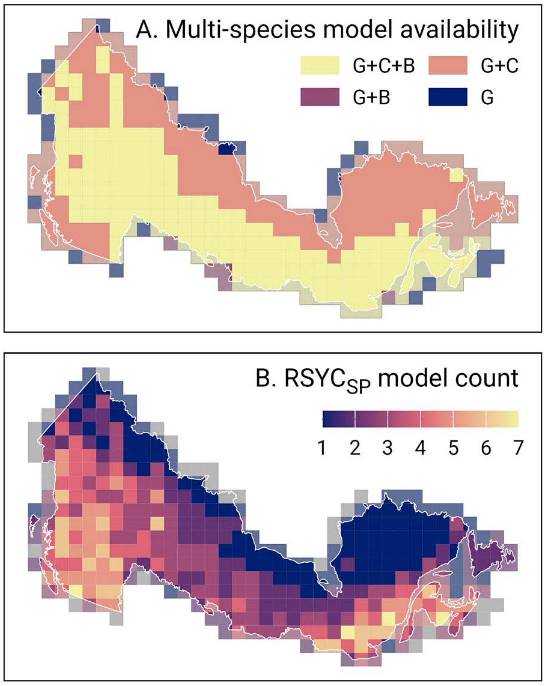

Figure 4. Availability of developed RSYC models. Panel A shows the availability of multi-species models, while panel B shows the number of species-specific models generated per tile. G—availability of generic models, B—availability of models for broadleaf species, C—availability of models for coniferous species.

models, with unique yield curves for a total of 308 tiles (Fig. 3). Model parameter estimates (Equation 2) are included in Table 1. A full list of the tile-level random parameters is provided in Supplementary materials S1.

The variance components (Table 1) revealed that most of the variability in AGB was attributed to differences at the tile (σ_i^2) and pixel (σ_{ij}^2) levels, with relatively low residual variance (σ^2_{error}), indicating that the hierarchical structure of the models effectively captured most of the variability in AGB. In all cases, conditional R^2 was markedly larger than marginal R^2, indicating that the fixed model parameters (i.e. parameters defining model form) explained only a small fraction of the variance, while the random effects (i.e. tile- and pixel-level variation) contributed substantially to the model's explanatory power. This result suggests that spatially structured variability has a much greater influence on model performance than the global shape of the yield curve, underscoring the importance of accounting for local conditions when modeling AGB dynamics.

Model evaluation results

Among the multispecies models, RSYC$_G$ had the largest proportion of tile-level validated models (30.4%) (Table 2, Supplementary material Fig. S3). These generic models achieved an R^2 of 0.36, with a bias of −10.34%, indicating a general tendency to underestimate AGB. RSYC$_C$ models had a slightly lower R^2 of 0.32, exhibited the lowest bias (2.06%) but highest RMSE% (50.38%) among all multispecies models. In contrast, RSYC$_B$ models had the highest bias (−20.70%) with an R^2 of 0.26. Note that all model evaluation results are based on weighted means, adjusted using the raking ratio method to account for imbalances in the distribution of validation plots. For comparison, results calculated without applying the weights are provided in the Supplementary material (Table S4).

The proportion of models validated using the MAGPlot data varied considerably across species-specific models, with the majority of validated models located in managed forests where field data were available. Some species had all developed models validated (e.g. yellow birch had five models and all were validated), while others had only a subset or none (e.g. Eastern white-cedar, red spruce, Tamarack). Overall, models for 13 out of 27 species were validated, with considerable variation in accuracy metrics. The highest accuracy was observed for models developed for black spruce and Jack pine. RSYC$_{SP}$ for black spruce had the lowest bias among all species-specific models at −3.46%, with a relative RMSE of 34.69% and R^2 of 0.39. Models for Jack pine had the highest R^2 (0.57), although a bias of −20.25% suggests a tendency to underestimate AGB. Conversely, several models exhibited low R^2 values along with high bias and RMSE. The largest biases were observed for trembling aspen, yellow birch, and Douglas-fir, with AGB underestimated by −27.99% for trembling aspen and overestimated by 28.93% for yellow birch. Models for white birch, lodgepole pine had the highest relative RMSE, reaching 49.68% and 52.31%, respectively.

The proportion of RSYC predictions that fell within the 95% confidence intervals of the mean field-measured AGB varied by model and age class (Fig. 6, solid blue lines). No consistent relationship was observed between the proportion of models within the 95% confidence interval and age. For some models, the proportion of predictions increased with age (e.g. Jack pine, lodgepole pine); while for others, it decreased (e.g. white birch). Overall, the proportion of models within the 95% confidence interval was equal or larger than 0.5 for all multi-species models and for 7 out of 13 of validated species-specific models. Among the multi-species models, RSYC$_C$ had the highest proportion (0.61), followed by RSYC$_B$ (0.54). Species-specific models generally had higher proportions of predictions within the 95% confidence interval of the mean field-measured AGB, with the highest values for white spruce (0.88). The lowest proportion was observed for eastern white-cedar (0.20).

The spatial variability of the tile-level validation results are shown in Fig. 7 (bias%) and Fig. 8 (RMSE%). Overall, the spatial distribution of both bias% and RMSE% did not reveal any strong regional patterns, with most tiles showing relatively consistent values. Variability in RMSE% was more uniform, with CV values ranging from 0.23 to 0.73. In contrast, the CV was greater for bias%, particularly for RSYC$_C$ and black spruce, which showed the highest CV values among all models (7.89 and 52.35, respectively).

For RSYC$_G$, a weak spatial trend was observed, with larger bias% values in Ontario compared to Quebec. A similar pattern was found for trembling aspen, with large negative bias% values in Ontario and smaller positive bias% values in Quebec. In western Canada (British Columbia and Yukon), bias% values were more variable for both RSYC$_G$ and, notably, RSYC$_C$, with large differences between neighboring tiles. RMSE% values were generally higher in western Canada for both RSYC$_G$ and RSYC$_C$.

Comparison to Ung et al.

Ung et al. (2009) yield curves were derived for 21 species (Supplementary material, Fig. S5). Validation against plot data indicated that the performance of the Ung2009 models was comparable or slightly lower than the RSYC models (Table 2, Fig. 6). The best-performing Ung2009 model was for trembling aspen, with an R^2 of 0.43, a bias of 6.98%, and an RMSE of 43.43%, outperforming the RSYC model for that species. However, this was

Table 1. Estimated coefficients (and their standard errors) for the fitted models by model type and species, as well as variance components of the random effects (σ^2_{i} and σ^2_{ij}).

	N	β_1	β_2	β_3	β_4	σ^2_{i}	σ^2_{ij}	σ^2_{error}	RMSE	$R^2_{marginal}$	$R^2_{conditional}$
Multispecies											
Generic	389	69.97 (2.09)	0.04923 (0.000049)	1.590 (0.002)	−0.00035 (0.000003)	1670.81	1032.13	77.21	8.67	0.04	0.97
Coniferous	355	75.36 (2.31)	0.04585 (0.000017)	1.639 (0.001)	0.00018 (0.000001)	1887.85	1890.68	104.64	10.18	0.02	0.98
Broadleaf	184	102.00 (1.73)	0.05768 (0.000046)	1.890 (0.002)	−0.00018 (0.000003)	552.38	974.98	113.12	10.55	0.14	0.92
Species-specific											
Black spruce	308	51.80 (1.30)	0.04511 (0.000062)	1.417 (0.002)	−0.00132 (0.000004)	518.80	424.59	68.25	8.15	0.09	0.94
Trembling aspen	151	107.85 (1.80)	0.05470 (0.000078)	1.894 (0.003)	0.00015 (0.000005)	488.63	1166.63	129.41	11.22	0.13	0.93
Jack pine	68	58.64 (3.08)	0.04905 (0.000149)	1.463 (0.005)	−0.00091 (0.000008)	644.83	354.38	74.08	8.49	0.08	0.94
White spruce	55	98.27 (5.25)	0.04990 (0.000114)	1.824 (0.005)	−0.00060 (0.000007)	1515.03	1062.86	112.23	10.46	0.09	0.96
Lodgepole pine	54	118.13 (5.01)	0.05266 (0.000109)	1.846 (0.004)	0.00023 (0.000006)	1352.46	2520.12	117.38	10.69	0.06	0.97
Balsam fir	52	79.15 (3.18)	0.05552 (0.000160)	1.541 (0.005)	−0.00010 (0.000008)	523.62	399.52	89.47	9.33	0.10	0.91
Subalpine fir	45	106.87 (6.63)	0.04298 (0.000105)	1.686 (0.004)	−0.00003 (0.000007)	1975.73	2969.13	107.87	10.25	0.06	0.98
White birch	42	87.16 (3.85)	0.06516 (0.000176)	1.838 (0.006)	−0.00042 (0.000008)	623.35	502.03	94.13	9.57	0.11	0.93
Sugar maple	28	124.90 (1.55)	0.05918 (0.000146)	2.027 (0.006)	−0.00022 (0.000008)	66.49	656.94	106.98	10.21	0.30	0.90
Western hemlock	27	192.88 (7.52)	0.05088 (0.000133)	1.690 (0.005)	−0.00006 (0.000006)	1522.49	5842.92	151.07	12.13	0.08	0.98
Engelmann spruce	25	153.42 (6.38)	0.04732 (0.000130)	1.890 (0.006)	0.00007 (0.000008)	1015.17	3475.42	139.13	11.64	0.10	0.97
Douglas-fir	16	205.01 (5.96)	0.05147 (0.000163)	1.963 (0.008)	0.00041 (0.000007)	561.36	8716.41	149.13	12.05	0.08	0.98
Mountain hemlock	16	135.64 (9.62)	0.05592 (0.000221)	1.860 (0.009)	0.00060 (0.000011)	1476.15	4257.02	134.93	11.46	0.05	0.97
Western redcedar	13	191.59 (3.69)	0.04926 (0.000181)	1.948 (0.009)	0.00017 (0.000010)	173.74	3354.86	151.49	12.15	0.17	0.96
Red maple	12	89.92 (7.06)	0.06042 (0.000308)	1.956 (0.012)	−0.00068 (0.000016)	597.48	1015.76	107.59	10.23	0.12	0.94
Eastern white pine	12	104.06 (2.22)	0.06571 (0.000285)	1.892 (0.011)	−0.00042 (0.000012)	58.50	418.90	92.68	9.50	0.23	0.87
Tamarack	10	26.78 (3.14)	0.04604 (0.000486)	1.374 (0.013)	−0.00551 (0.000032)	98.31	214.70	49.71	6.95	0.23	0.94
Red spruce	7	124.10 (6.35)	0.04882 (0.000326)	1.754 (0.012)	0.00164 (0.000022)	280.46	823.05	117.30	10.69	0.16	0.88
Largetooth aspen	6	104.53 (4.82)	0.06349 (0.000503)	2.008 (0.021)	0.00032 (0.000025)	138.27	696.73	124.88	11.02	0.14	0.87
Yellow birch	5	115.17 (1.63)	0.06926 (0.000425)	2.204 (0.021)	−0.00132 (0.000016)	12.72	401.29	90.22	9.38	0.34	0.89
Amabilis fir	4	228.03 (11.52)	0.04563 (0.000271)	1.802 (0.012)	0.00058 (0.000015)	523.90	9095.89	146.33	11.94	0.12	0.98
Yellow-cedar	4	163.78 (10.94)	0.05968 (0.000468)	1.893 (0.019)	0.00071 (0.000023)	473.83	4388.38	141.55	11.74	0.09	0.97
Eastern white-cedar	3	100.21 (5.30)	0.05930 (0.000830)	1.509 (0.027)	0.00044 (0.000033)	83.67	348.13	100.04	9.87	0.10	0.81
Red pine	2	86.69 (0.99)	0.06773 (0.001286)	1.583 (0.049)	−0.00177 (0.000034)	1.53	245.47	83.71	9.03	0.23	0.84
Black ash	1	116.49 (1.00)	0.04730 (0.000742)	1.679 (0.025)	−0.00014 (0.000051)	0.00	641.17	106.40	10.17	0.34	0.89
Western larch	1	234.27 (2.18)	0.05472 (0.000725)	2.320 (0.048)	0.00095 (0.000033)	0.01	4468.82	139.67	11.67	0.09	0.96
Balsam poplar	1	133.37 (1.88)	0.04027 (0.000853)	1.362 (0.027)	0.00086 (0.000056)	0.01	3503.66	126.21	11.08	0.09	0.96

Conditional R^2 calculated at level i. N—total number of tile-level random effects, RMSE—root mean square error. For all models, the resulting parameter estimates were significant (P-value <.05).

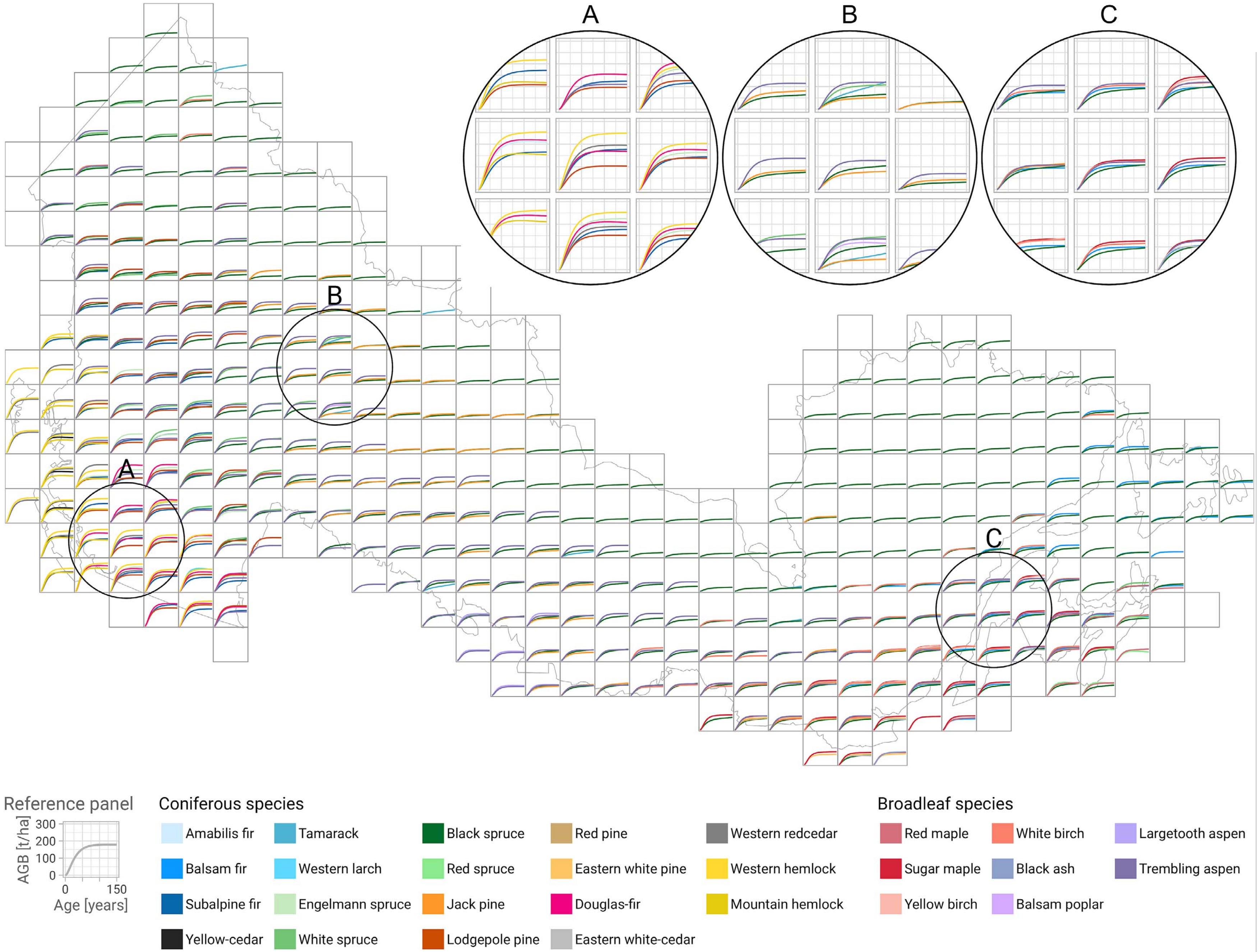

Figure 5. National map showing 150 × 150 km analysis tiles where species-specific RSYC models were developed. The reference panel illustrates the range of age (x-axis) and AGB (y-axis) values plotted on each tile. Three circular insets (A, B, and C) provide zoomed-in views of selected regions.

Table 2. Strata-level validation results for RSYC and Ung et al. 2009.

		RSYC					Ung et al. 2009				
	N validated	R^2	bias	bias%	RMSE	RMSE%	R^2	bias	bias%	RMSE	RMSE%
Multispecies models											
Generic	117 (30.4%)	0.36	−10.34	−11.31	41.53	45.41					
Coniferous	84 (23.7%)	0.32	1.68	2.06	41.03	50.38					
Broadleaf	28 (15.2%)	0.26	−22.23	−20.70	51.51	47.97					
Species-specific models											
Black spruce	50 (16.2%)	0.39	−2.27	−3.46	22.79	34.69	0.29	11.71	17.67	35.05	52.89
Trembling aspen	23 (15.2%)	0.28	−30.59	−27.99	53.31	48.76	0.43	7.68	6.98	47.77	43.43
Jack pine	13 (19.1%)	0.57	−18.59	−20.25	32.51	35.41	0.41	52.47	57.00	103.82	112.79
White spruce	3 (5.5%)	0.05	−10.33	−14.18	29.96	41.13	0.09	19.51	29.49	43.09	65.14
Lodgepole pine	12 (22.2%)	0.37	2.39	2.60	48.15	52.31	0.53	73.48	78.10	96.76	102.84
Balsam fir	20 (38.5%)	0.31	9.35	14.46	27.09	41.90	0.19	5.95	8.85	34.54	51.40
White birch	16 (38.1%)	0.28	16.80	23.74	35.16	49.68	0.30	11.62	16.54	29.58	42.10
Sugar maple	14 (50%)	0.08	−12.31	−10.81	49.54	43.50					
Douglas-fir	4 (25%)	0.36	28.60	21.29	64.15	47.74	0.31	−1.92	−1.43	58.04	43.20
Red maple	1 (8.3%)	0.48	−3.91	−4.26	30.61	33.39	0.43	10.23	11.16	34.58	37.72
Eastern white pine	5 (41.7%)	0.34	−4.39	−4.27	41.20	40.07	0.39	0.25	0.23	46.52	43.40
Yellow birch	5 (100%)	0.46	22.58	28.93	35.83	45.90	0.44	16.72	21.43	33.85	43.36
Eastern white-cedar	1 (33.3%)	0.16	16.37	20.99	37.32	47.85	0.09	56.79	72.80	67.30	86.28

Note: Ung et al. 2009 does not include multi-species models.

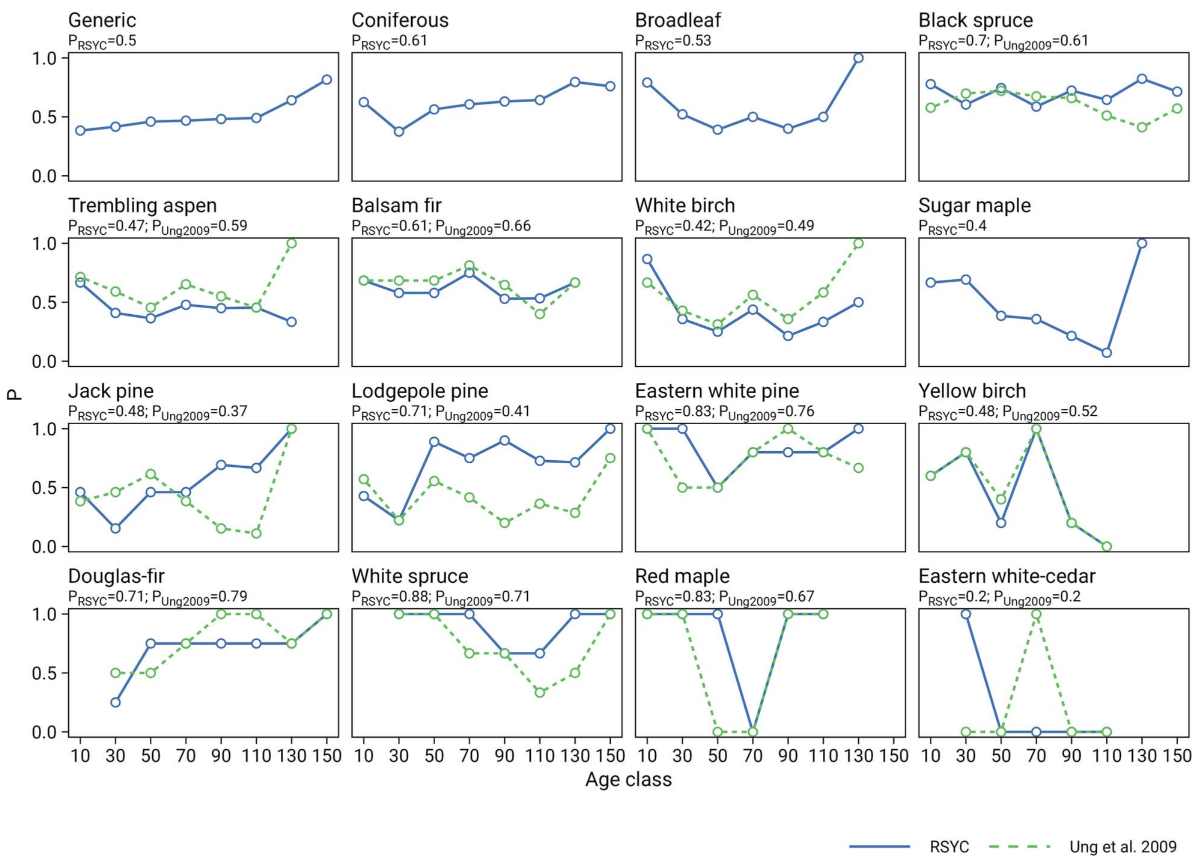

Figure 6. Proportion of RSYC and Ung et al. 2009 (Ung2009) predictions that fell within the 95% confidence intervals of the mean field-measured AGB for each age class.

not consistent across species, as the Ung2009 models for Jack pine, lodgepole pine, and eastern white-cedar showed significantly higher bias and RMSE compared to RSYC (Table 2). The proportion of Ung2009 estimates falling within the 95% confidence interval of field-measured AGB was generally comparable to RSYC models, except for lodgepole pine, where it was markedly lower (Fig. 6).

Comparison between the two model sets (Supplementary materials Fig. S5) revealed that while some species, such as black spruce and yellow birch exhibited similar yield curve shapes

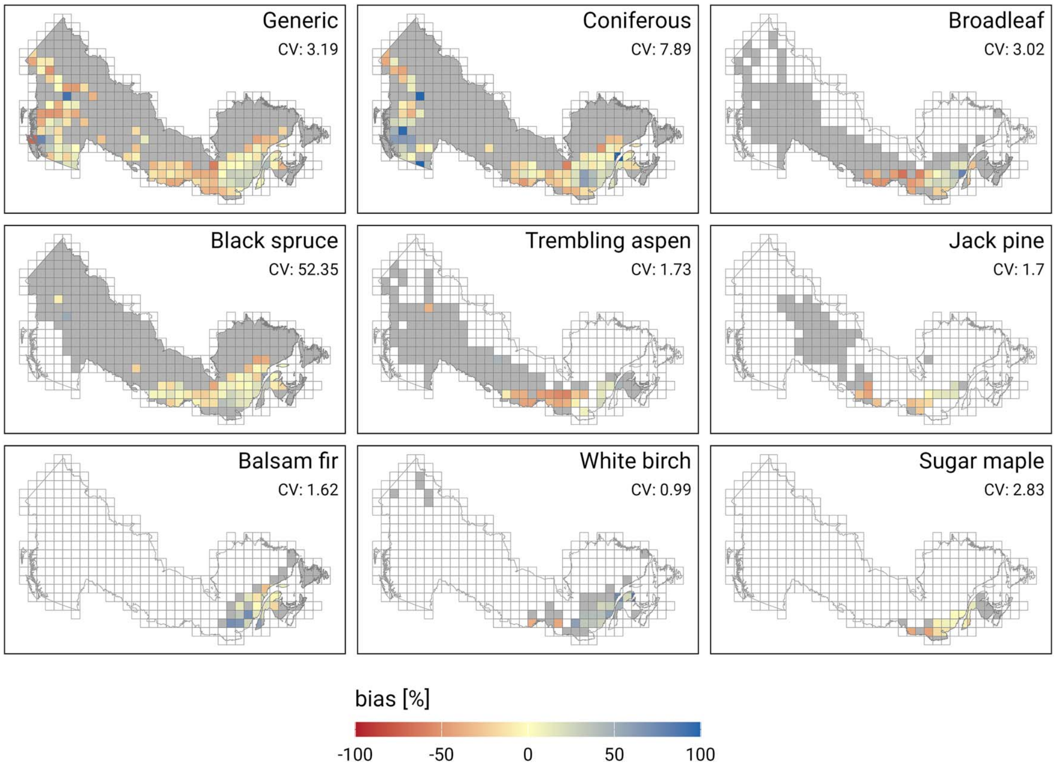

Figure 7. Tile-level relative bias by model type. Grey shading indicates whether the model was developed for particular tile. Note that in addition to $RSYC_G$ (Generic), $RCYC_C$ (Coniferous), and $RSYC_B$ (Broadleaf), the six species-specific models validated on the largest number of tiles are shown. See Supplementary material S1 for the complete information on validation results. CV—coefficient of variation.

between models, the majority showed substantial discrepancies. RSYC yield curves generally predicted lower AGB than Ung2009. For Ung2009, bias values were positive for most species, with the highest bias observed for lodgepole pine (78.10%). The largest overlap between models occurred for white spruce, black spruce, subalpine fir, and tamarack.

Figure S5 highlights stronger agreement between the two models for black spruce, white spruce, subalpine fir, tamarack, red spruce, and yellow birch. In contrast, for species such as Jack pine and western hemlock, Ung2009 predicted unrealistically high AGB values, exceeding 800 t/ha in the case of western hemlock. These discrepancies suggest potential overestimation in the Ung2009 model for certain species.

Discussion

Aboveground biomass yield curves for treed areas in Canada

We generated a database of yield curves that constitutes a large, spatially representative set of AGB equations for Canada. These RSYC were developed for all of Canada's forested ecozones, and represent 27 individual species, along with three multispecies models: one for coniferous species, one for broadleaf species, and one generic, species-independent model. Each model represents the average AGB density trajectory (t/ha) over stand age within a 150 × 150 km tile, thereby capturing broad-scale spatial variation in productivity and environmental conditions. The models were constructed using Landsat time-series-derived forest age, species, and AGB, adapting the methodology outlined in Tompalski et al. (2024). Key advancements include a more flexible nonlinear mixed-effects model formulation that separately parameterizes the early growth and mature phases, improving the representation of rapid biomass accumulation in young stands and potential decline in older stands. Tile-specific random effects were incorporated to capture local variability across Canada's diverse forest regions and to increase model coverage, as models could be developed even when the available samples within a tile did not span the full age range. This approach contrasts with fitting separate models for each tile, which would have resulted in fewer yield curves due to sample size limitations. Species coverage was expanded considerably, resulting in a more comprehensive set of yield curves applicable to both managed and unmanaged forests.

A primary objective of these national yield curves is to facilitate large-area assessments of forest growth and carbon dynamics using a consistent, nationally harmonized framework, aligned with the research priorities identified in Canada's updated Carbon Blueprint (Smyth et al. 2024). Unlike traditional growth and yield models, which are often developed independently by provincial or territorial agencies (Metsaranta et al. 2024), the RSYC approach applies the same methods and input data across all jurisdictions, ecozones, and species. This methodological consistency allows for direct comparisons among regions and supports integrated national-scale assessments. Estimates derived from such a harmonized model are also less likely to be biased than those generated by assembling outputs from a mix of jurisdiction-specific models, which often differ in structure, assumptions, and data sources (British Columbia Ministry of Forests 2009, Huang et al. 2009, Pretzsch 2009).

The national yield curves also fill important model availability gaps, particularly in the unmanaged forests of northern Canada

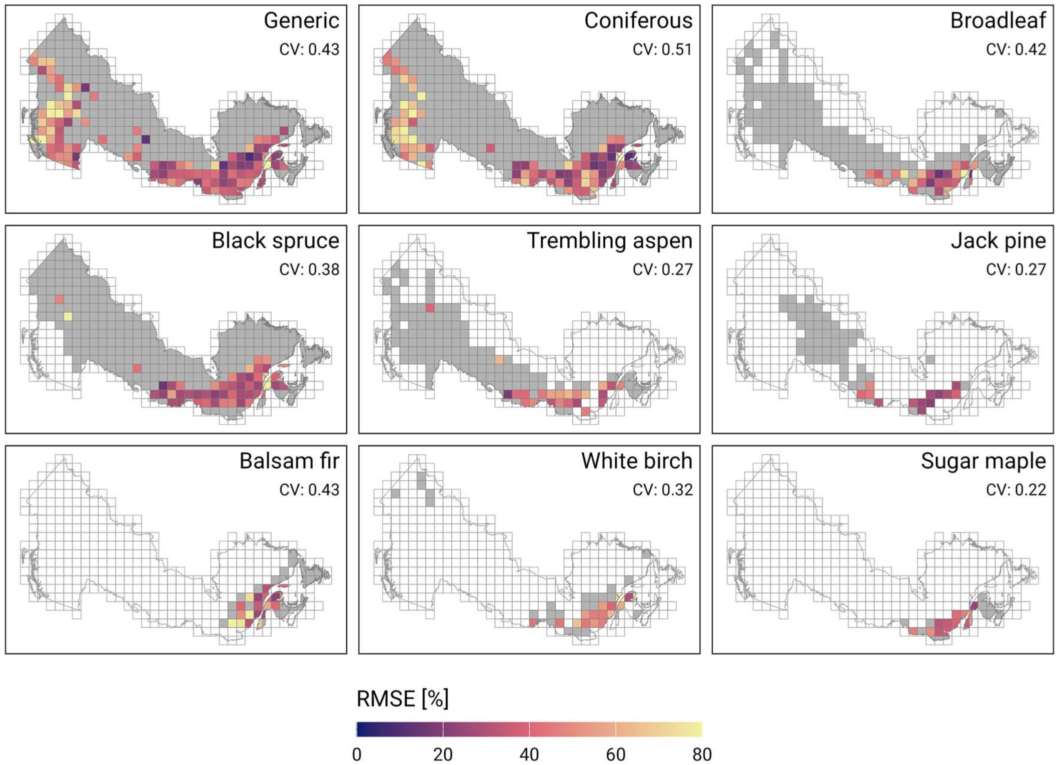

Figure 8. Tile-level relative RMSE by model type. Grey shading indicates whether the model was developed for a particular tile. Note that in addition to RSYC$_G$ (Generic), RCYC$_C$ (Coniferous), and RSYC$_B$ (Broadleaf), the six species-specific models validated on the largest number of tiles are shown. See Supplementary material S1 for the complete information on validation results. CV—coefficient of variation.

(Fig. 1) where traditional growth and yield data are scarce (Boisvenue and White 2019). These low-productivity stands have historically been excluded from empirical yield research, yet they represent a significant component of Canada's forested landscape and are important for understanding national-scale carbon dynamics (Nabuurs et al. 2023). Remote sensing offers a practical and scalable solution for capturing biomass accumulation in these underrepresented regions (Matasci et al. 2018a, Nguyen et al. 2020, Morin-Bernard et al. 2023, Guindon et al. 2024, Hunka et al. 2024). In doing so, the RSYC models produce information that is critical for national carbon monitoring and reporting, and also supports a range of forest management applications in areas where field-based data are limited and conventional planning tools are unavailable.

Compared to traditional modeling approaches, the RSYC models are based on a substantially larger input dataset, allowing for the development of yield curves with broader spatial and temporal coverage. The Landsat record enables tracking of AGB dynamics over several decades, with over four decades of data available at 30-m spatial resolution (Wulder et al. 2022). This temporal depth provides a unique opportunity to reconstruct growth trajectories without relying on long-term permanent sample plots, which are limited in number and geographic coverage. While temporary sample plots are more widespread, they lack the repeat measurements necessary to model change over time.

At the same time, the RSYC approach is not without challenges. The input variables used to construct the models (forest age, species composition, and AGB) are themselves model-based estimates that have inherent uncertainties (Matasci et al. 2018b, Maltman et al. 2023, Hermosilla et al. 2024). These uncertainties can lead to unrealistic AGB values at certain stand ages, such as anomalous high estimates in young stands or unlikely growth trajectories, requiring initial filtering to remove anomalous data (Tompalski et al. 2024). AGB estimates derived from remote sensing are particularly prone to overestimating lower biomass values (primarily affecting young forests) and underestimating higher values in older stands (Lefsky et al. 2002, Li et al. 2020, Ahmad et al. 2021). Our filtering approach was effective at identifying and removing overestimated AGB values in young stands, where age estimates are relatively accurate and growth patterns well understood, but underestimation in older stands may still persist.

While the RSYC framework offers clear advantages in terms of national consistency and the ability to generate yield curves in data-sparse regions, it is important to recognize its differences from traditional plot-based models, which are often developed using direct, repeated field measurements. In well-sampled and intensively managed forests, such plot-based models may achieve higher accuracy. In contrast, RSYC models rely on model-derived inputs and therefore inherit the uncertainties associated with those inputs. This trade-off means that while RSYC models enable national-scale coverage, their predictions may be less accurate or have greater uncertainty associated with them when compared to models based on high-quality, ground-measured data.

Model evaluation results

The evaluation of a forest growth model is a complex, multifaceted task that extends beyond the mere assessment of model accuracy (Pretzsch 2009). The RSYC models were evaluated following best practices for growth and yield model validation, as outlined by Vanclay and Skovsgaard (1997). Most importantly, the evaluation was based on an independent reference plot dataset—the plot data used for model evaluation were not used in the development of the remote sensing-derived inputs. This independence helped ensure that the validation results reflect true predictive performance of the developed models (Duncanson et al. 2021).

Vanclay and Skovsgaard (1997) emphasize that model evaluation should extend beyond empirical fit to include an assessment of biological realism. In the RSYC models, biological realism was supported through the choice of a flexible four-parameter model form, which allows two inflection points in the biomass trajectory. The first inflection point captures the rapid biomass accumulation typical of young stands, while the second inflection point enables a plateau or decline in older stands (e.g. yield curves developed for red spruce or yellow cedar), reflecting processes such as increased mortality or stand breakup (Larson et al. 2015). This model form represented an improvement over the previously used three-parameter form (Tompalski et al. 2024) and allowed the yield curves to depict biologically plausible patterns (Weiskittel et al. 2011). In addition to empirical validation, each developed yield curve was also visually assessed to ensure that the predicted AGB trajectories were logical and biologically consistent.

Model evaluation was conducted at both the tile and strata levels. Strata-level validation provided an overall assessment of model performance across broad ecological divisions, while tile-level validation offered insights into regional variability and localized model behavior. Combining these two levels of analysis allowed a more comprehensive understanding of model strengths and limitations. The validation results showed considerable variation in model performance across species (for species-specific models) and species groups (for multi-species models), underscoring the influence of species composition on model performance. Some models demonstrated strong agreement with field data, particularly those developed for black spruce, lodgepole pine, red maple, eastern white pine, and the multi-species coniferous model ($RSYC_C$). These models exhibited low bias overall and relatively small errors. Other models, however, showed substantial bias and higher RMSE values (both absolute and relative), suggesting that species-specific growth patterns, allometric relationships, or environmental variability may be more challenging to capture for certain species (e.g. trembling aspen, sugar maple).

The proportion of RSYC predictions falling within the 95% confidence intervals of field-measured AGB also varied by species and age class. While some species showed consistently high agreement across age classes, others exhibited patterns of declining or increasing model performance with age, further indicating that growth trajectories are not uniformly captured across all species.

Tile-level validation revealed larger variability in bias compared to RMSE, suggesting that spatial variation in model performance is primarily driven by systematic over or underestimation rather than random errors. This variation may be partly explained by differences in the quality of inventory data, forest composition, disturbance histories, or environmental gradients across tiles, but it may also indicate that growth responses are highly localized, underscoring the importance of including random effects in the model. Examination of the tile-level results (Supplementary materials, Fig. S2) showed that for some tiles, the reference values deviated markedly from expected ranges for given stand ages (e.g. high AGB at very young age). Such deviations suggest that some inconsistencies or errors in the reference data from the ground could have influenced the observed validation patterns.

Another useful test of the model's biological realism is to compare the outcomes of alternative models for the same area (Weiskittel et al. 2011). To put RSYC model performance into perspective, we also validated one of the existing yield curve models currently widely used in Canada's national carbon accounting framework, specifically the model developed by Ung et al. (2009). We applied the same validation approach to both RSYC and Ung2009 models, enabling a direct comparison between the two models. Depending on the species, the RSYC models showed similar or slightly improved model performance relative to Ung2009. The main difference was bias: Ung2009 tended to systematically overestimate AGB, while RSYC models showed lower and less systematic bias. Although traditional model performance metrics such as RMSE and R^2 provide useful information on overall fit and variability explained, we argue that in this context, minimizing bias is particularly important—especially when models are used to support carbon accounting and forest resource management and planning.

Limitations and opportunities

While the national yield curves developed using the RSYC approach offer a consistent and spatially comprehensive framework for estimating AGB across Canada, several limitations should be acknowledged. Model evaluation is restricted to areas where plot data are available, which are predominantly managed forests in more southern regions of the country. As a result, we can only assume that the model performs similarly in unmanaged forests. This assumption may be reasonable, as unmanaged stands are often simpler in structure, grow more slowly, consist of fewer species, and tend to be more open, but without sufficient ground-based data, we are currently unable to confirm model performance in these regions. Establishing permanent sample plots in unmanaged forests would help address this limitation by providing essential calibration and validation data for remote sensing-based models. The validation data used MAGPlot (National Forest Inventory 2023), which is a compilation of multiple jurisdictional datasets collected under different standards and protocols. Despite ongoing efforts to standardize the database, inconsistencies likely persist. In particular, forest age is often not measured directly at the plot level but is instead assigned from the associated forest inventory polygon. These stand-level ages are typically derived from photointerpretation (e.g. OMNR 2009, QC MFFP 2015, BC MFLNRORD 2020), which introduces additional uncertainty, especially in older stands.

The performance of the RSYC models also depends on the quality of the remote sensing-derived inputs, with each input (AGB, age, species) contributing differently to overall uncertainty. Biomass estimates derived from satellite imagery, such as Landsat, are known to underestimate AGB in older stands (Lefsky et al. 2002, Ahmad et al. 2021), which likely explains the predominantly negative bias observed in the RSYC models. In contrast, age estimates are most accurate for stands that were disturbed within the Landsat record wherein the timing of disturbance can be identified with high precision (typically within ± year) leading to increased confidence in the early portions of the yield curves where the majority of biomass accumulation occurs. While errors in species classification are possible, this issue was mitigated by including only pixels where the dominant species remained consistent throughout the time series, thereby reducing the likelihood of misclassification influencing the modeled trajectories.

Despite the challenges, the national RSYC models represent a significant advancement in forest growth modeling for large and diverse forest areas, particularly in data-sparse regions. The developed models provide a robust foundation for continued improvement with the availability of additional ground data and higher-resolution remote sensing inputs. The models can be recalibrated using longer time series and as additional datasets are integrated, including ALS or alternative satellite-derived products. Finally, the modeling framework is flexible, allowing the development of models at various spatial units (e.g. ecoregions) or for different species groupings.

Conclusions

Yield curves are essential for understanding forest growth and informing carbon accounting, especially in regions where traditional inventory data are limited. In this study, we developed a set of AGB yield curves using remote sensing-derived time series of forest age, species composition, and AGB across Canada's forested ecozones. A nonlinear mixed-effects modeling approach was applied at a 150×150 km tile level, capturing local variability through nested random effects. Validation against harmonized field plot data showed that the RSYC models generally outperformed or matched the performance of existing national models. While model performance varied by species, and some limitations remain due to input uncertainties and limited field data in unmanaged forests, the RSYC models offer a consistent, scalable framework to support national forest monitoring and carbon estimation. Users can select from 27 species-specific, three multi-species, and one generic model. An R package has been developed to aid users in model selection and to facilitate access.

Acknowledgements

We thank Mathieu Fortin of the Canadian Forest Service for assistance with model evaluation. We also thank the editor and three anonymous reviewers for their constructive comments during revision.

Author contributions

Piotr Tompalski (Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Validation, Visualization, Writing—original draft, Writing—review & editing), Txomin Hermosilla (Conceptualization, Data curation, Investigation, Methodology, Writing—review & editing), Sharad Kumar Baral (Conceptualization, Methodology, Validation, Writing—review & editing), Michael A Wulder (Conceptualization, Methodology, Writing—review & editing), and Joanne White (Conceptualization, Methodology, Writing—review & editing)

Supplementary data

Supplementary data are available at *Forestry* online.

Conflict of interest

None declared.

Funding

This work was supported by the Canadian Forest Service of Natural Resources Canada.

Data availability

The model coefficients generated in this study for Canada's forested ecosystems are open access and are available in the Supplementary materials (S1). We have also developed an R package available at https://github.com/ptompalski/RSYC which allows users easy access to the developed models.

References

Abercrombie SP, Friedl MA. Improving the consistency of multi-temporal land cover maps using a hidden Markov model. *IEEE Trans Geosci Remote Sens* 2016;**54**:703–13. https://doi.org/10.1109/TGRS.2015.2463689.

Ahmad A, Gilani H, Ahmad SR. Forest aboveground biomass estimation and mapping through high-resolution optical satellite imagery—a literature review. *Forests* 2021;**12**:914. https://doi.org/10.3390/f12070914.

Anderson J, Martin ME, Smith M-L. *et al.* The use of waveform lidar to measure northern temperate mixed conifer and deciduous forest structure in New Hampshire. *Remote Sens Environ* 2006;**105**:248–61. https://doi.org/10.1016/j.rse.2006.07.001.

Birdsey R, Pan Y. Trends in management of the world's forests and impacts on carbon stocks. *Forest Ecology and Management, Carbon, water and nutrient cycling in managed forests* 2015;**355**:83–90. https://doi.org/10.1016/j.foreco.2015.04.031.

Boisvenue C, White JC. Information needs of next-generation Forest carbon models: opportunities for remote sensing science. *Remote Sens* 2019;**11**:463. https://doi.org/10.3390/rs11040463.

Boudewyn PA, Song X, Magnussen S. *et al.* Model based volume-to-biomass conversion for forested and vegetated land in Canada, forestry. In: *Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Information Report BC-X-411, Victoria*. Canada, 2007, 112 p.

British Columbia Ministry of Forests; Lands; Natural Resource Operations and Rural Development (BC MFLNRORD). *Variable Density Yield Projection*, Vol. **1**. VDYP7 Overview, 2019. Victoria, BC, Canada.

British Columbia Ministry of Forests; Lands; Natural Resource Operations and Rural Development (BC MFLNRORD). Vegetation resources inventory photo interpretation procedures. 2020. Retrieved from: https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/nr-laws-policy/risc/vri_photo_interpretation_procedures_v36_2020.pdf.

Canadian CASFRI Working Group. Common attribute schema for Forest resource inventories. 2025; CASFRI version 1.1.0.

Duncanson L, Armston J, Disney M. *et al. Aboveground Woody Biomass Product Validation Good Practices Protocol* 2021, p. 236. https://doi.org/10.5067/DOC/CEOSWGCV/LPV/AGB.001.

Farrar JL. Trees in Canada: Revised centenary edition. In: *Fitzhenry & Whiteside [u.a.]*, 2017 Ottawa: Markham, Ontario.

Fassnacht FE, White JC, Wulder MA. *et al*. Remote sensing in forestry: current challenges, considerations and directions. *Forestry: An International Journal of Forest Research*cpad024 2023;**97**:11–37. https://doi.org/10.1093/forestry/cpad024.

Fortin M, Lavoie J-F. Reconciling individual-based forest growth models with landscape-level studies through a meta-modelling approach. *Can J For Res* 2022;**52**:1140–53. https://doi.org/10.1139/cjfr-2022-0002.

Guindon L, Manka F, Correia DLP. *et al.* A new approach for spatializing the CAnadian National Forest Inventory (SCANFI) using Landsat dense time series. *Can J For Res* 2024;**54**:793–815. https://doi.org/10.1139/cjfr-2023-0118.

Hermosilla T, Bastyr A, Coops NC. *et al.* Mapping the presence and distribution of tree species in Canada's forested ecosystems. *Remote Sens Environ* 2022a;**282**:113276. https://doi.org/10.1016/j.rse.2022.113276.

Hermosilla T, Wulder MA, White JC. *et al.* Regional detection, characterization, and attribution of annual forest change from 1984 to 2012 using Landsat-derived time-series metrics. *Remote Sens Environ* 2015a;**170**:121–32. https://doi.org/10.1016/j.rse.2015.09.004.

Hermosilla T, Wulder MA, White JC. *et al.* Mass data processing of time series Landsat imagery: pixels to data products for forest monitoring. *International Journal of Digital Earth* 2016;**9**:1035–54. https://doi.org/10.1080/17538947.2016.1187673.

Hermosilla T, Wulder MA, White JC. *et al.* Prevalence of multiple forest disturbances and impact on vegetation regrowth from interannual Landsat time series (1985–2015). *Remote Sens Environ* 2019;**233**:111403. https://doi.org/10.1016/j.rse.2019.111403.

Hermosilla T, Wulder MA, White JC. *et al.* Land cover classification in an era of big and open data: optimizing localized implementation and training data selection to improve mapping outcomes. *Remote Sens Environ* 2022b;**268**:112780. https://doi.org/10.1016/j.rse.2021.112780.

Hermosilla T, Wulder MA, White JC. *et al.* Characterizing long-term tree species dynamics in Canada's forested ecosystems using annual time series remote sensing data. *For Ecol Manag* 2024;**572**:122313. https://doi.org/10.1016/j.foreco.2024.122313.

Hermosilla T, Wulder MA, White JC. *et al.* An integrated Landsat time series protocol for change detection and generation of annual gap-free surface reflectance composites. *Remote Sens Environ* 2015b;**158**:220–34. https://doi.org/10.1016/j.rse.2014.11.005.

Huang S, Meng SX, Yang Y. A growth and yield projection system (GYPSY) for natural and post-harvest stands in Alberta. *Forestry Division, Alberta Sustainable Resource Development Technical Report* 2009;**T/216**:1–22.

Hunka N, Duncanson L, Armston J. *et al.* Intergovernmental panel on climate change (IPCC) tier 1 forest biomass estimates from earth observation. *Sci Data* 2024;**11**:1127. https://doi.org/10.1038/s41597-024-03930-9.

Kim H, Kim Y-H, Kim R. *et al.* Reviews of forest carbon dynamics models that use empirical yield curves: CBM-CFS3, CO2FIX, CASMOFOR, EFISCEN. *For Sci Technol* 2015;**11**:212–22. https://doi.org/10.1080/21580103.2014.987325.

Kurz WA, Apps M, Banfield E. *et al.* Forest carbon accounting at the operational scale. *For Chron* 2002;**78**:672–9. https://doi.org/10.5558/tfc78672-5.

Kurz WA, Dymond CC, White TM. *et al.* CBM-CFS3: a model of carbon-dynamics in forestry and land-use change implementing IPCC standards. *Ecol Model* 2009;**220**:480–504. https://doi.org/10.1016/j.ecolmodel.2008.10.018.

Kurz WA, Shaw CH, Boisvenue C. *et al.* Carbon in Canada's boreal forest — a synthesis. *Environ Rev* 2013;**21**:260–92. https://doi.org/10.1139/er-2013-0041.

Lambert MC, Ung CH, Raulier F. Canadian national tree aboveground biomass equations. *Can J For Res* 2005;**35**:1996–2018. https://doi.org/10.1139/x05-112.

Larson AJ, Lutz JA, Donato DC. *et al.* Spatial aspects of tree mortality strongly differ between young and old-growth forests. *Ecology* 2015;**96**:2855–61. https://doi.org/10.1890/15-0628.1.

Lefsky MA, Cohen WB, Harding DJ. *et al.* Lidar remote sensing of above-ground biomass in three biomes. *Glob Ecol Biogeogr* 2002;**11**:393–9. https://doi.org/10.1046/j.1466-822x.2002.00303.x.

Li Y, Li M, Li C. *et al.* Forest aboveground biomass estimation using Landsat 8 and sentinel-1A data with machine learning algorithms. *Sci Rep* 2020;**10**:9952. https://doi.org/10.1038/s41598-020-67024-3.

Magnussen S, Andersen H-E, Mundhenk P. A second look at endogenous poststratification. *Forensic Sci* 2015;**61**:624–34. https://doi.org/10.5849/forsci.14-183.

Maltman JC, Hermosilla T, Wulder MA. *et al.* Estimating and mapping forest age across Canada's forested ecosystems. *Remote Sens Environ* 2023;**290**:113529. https://doi.org/10.1016/j.rse.2023.113529.

Masera OR, Garza-Caligaris JF, Kanninen M. *et al.* Modeling carbon sequestration in afforestation, agroforestry and forest management projects: the CO2FIX V.2 approach. *Ecol Model* 2003;**164**:177–99. https://doi.org/10.1016/S0304-3800(02)00419-2.

Matasci G, Hermosilla T, Wulder MA. *et al.* Three decades of forest structural dynamics over Canada's forested ecosystems using Landsat time-series and lidar plots. *Remote Sens Environ* 2018a;**216**:697–714. https://doi.org/10.1016/j.rse.2018.07.024.

Matasci G, Hermosilla T, Wulder MA. *et al.* Large-area mapping of Canadian boreal forest cover, height, biomass and other structural attributes using Landsat composites and lidar plots. *Remote Sens Environ* 2018b;**209**:90–106. https://doi.org/10.1016/j.rse.2017.12.020.

Metsaranta JM, Fortin M, White JC. *et al.* Climate sensitive growth and yield models in Canadian forestry: challenges and opportunities. *For Chron* 2024;**100**:88–106. https://doi.org/10.5558/tfc2024-005.

Morin-Bernard A, Coops NC, White JC. *et al.* Predicting net growth rates in boreal forests using Landsat time series and permanent sample plot data. *Forestry: An International Journal of Forest Research*, 2023;cpad055. https://doi.org/10.1093/forestry/cpad055.

Nabuurs G-J, Ciais P, Grassi G. *et al.* Reporting carbon fluxes from unmanaged forest. *Commun Earth Environ* 2023;**4**:1–4. https://doi.org/10.1038/s43247-023-01005-y.

Nakagawa S, Schielzeth H. A general and simple method for obtaining R2 from generalized linear mixed-effects models. *Methods Ecol Evol* 2013;**4**:133–42. https://doi.org/10.1111/j.2041-210x.2012.00261.x.

National Forest Inventory. Multi-agency ground plot (MAGPlot) database – standards for forest ground plot data harmonization. 2023; version 1.0 [WWW Document]. URL http://nfi.nfis.org (accessed 3.1.22).

Natural Resources Canada 2025. *The State of Canada's Forests: Annual Report*, 2024.

Nguyen T, Jones S, Soto-Berelov M. *et al.* Landsat time-series for estimating Forest aboveground biomass and its dynamics across space and time: a review. *Remote Sens* 2020;**12**:98. https://doi.org/10.3390/rs12010098.

Ontario Ministry of Natural Resources (OMNR). Forest resources inventory technical specifications [WWW document]. 2009. https://www.ontario.ca/files/2025-05/fim-tech-spec-forest-resources-inventory.pdf

Packalen T, Sallnaes P, Sirkia S. *et al.* The European forestry dynamics model: concept, design and results of first case studies. 2014. Publications Office of the European Union, EUR 27004. https://doi.org/10.2788/153990

Pan Y, Birdsey RA, Phillips OL. *et al.* The enduring world forest carbon sink. *Nature* 2024;**631**:563–9. https://doi.org/10.1038/s41586-024-07602-x.

Pinheiro J, Bates D, R Core Team. Nlme: linear and nonlinear mixed effects models (manual). 2023, R package version 3.1-164, https://CRAN.R-project.org/package=nlme.

Pretzsch H. *Forest Dynamics, Growth and Yield*. Berlin Heidelberg, Berlin, Heidelberg: Springer, 2009, 1–39. https://doi.org/10.1007/978-3-540-88307-4_1.

Québec Ministère des Forêts de la Faune et des Parcs (QC MFFP). Norme de photo-interprétation - Quatrième inventaire écoforestier du Québec méridional. 2015. Retrieved from: https://mffp.gouv.qc.ca/publications/forets/connaissances/Norme-photo-interpretation.pdf.

R Core Team. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing, 2024.

Richards GP, Evans DMW. Development of a carbon accounting model (FullCAM Vers. 1.0) for the Australian continent. *Aust For* 2004;**67**:277–83. https://doi.org/10.1080/00049158.2004.10674947.

Schelhaas M-J, Eggers J, Lindner M. *et al. Model Documentation for the European Forest Information Scenario Model (EFISCEN 3.1. 3)*. Alterra report 1559 and EFI technical report 26. Alterra and European Forest Institute, Wageningen and Joensuu, p. 118.

Shifley SR, He HS, Lischke H. *et al.* The past and future of modeling forest dynamics: from growth and yield curves to forest landscape models. *Landsc Ecol* 2017;**32**:1307–25. https://doi.org/10.1007/s10980-017-0540-9.

Smyth C, Metsaranta J, Tompalski P. *et al.* 10-year progress on Forest carbon research in Canada. *Environ Rev* 2024;**32**:611–37. https://doi.org/10.1139/er-2024-0001.

Stinson G, Thandi G, Aitkin D. *et al.* A new approach for mapping forest management areas in Canada. *For Chron* 2019;**95**:101–12. https://doi.org/10.5558/tfc2019-017.

Tompalski P, Wulder MA, White JC. *et al.* Developing aboveground biomass yield curves for dominant boreal tree species from time series remote sensing data. *For Ecol Manag* 2024;**561**:121894. https://doi.org/10.1016/j.foreco.2024.121894.

Ung C-H, Bernier P, Guo X-J. Canadian national biomass equations: new parameter estimates that include British Columbia data. *Can J For Res* 2008;**38**:1123–32. https://doi.org/10.1139/X07-224.

Ung C-H, Bernier PY, Guo XJ. *et al.* A simple growth and yield model for assessing changes in standing volume across Canada's forests. *For Chron* 2009;**85**:57–64. https://doi.org/10.5558/tfc85057-1.

Vanclay JK, Skovsgaard JP. Evaluating forest growth models. *Ecological Modelling, Design, Performance and Evaluation of Models for Forest Stand Dynamics* 1997;**98**:1–12. https://doi.org/10.1016/S0304-3800(96)01932-1.

Wang T, Hamann A, Spittlehouse D. *et al.* Locally downscaled and spatially customizable climate data for historical and future periods for North America. *PLoS One* 2016;**11**:e0156720. https://doi.org/10.1371/journal.pone.0156720.

Weiskittel AR, Hann DW, Kershaw JA. *et al. Forest Growth and Yield Modeling*. Forest Growth and Yield Modeling: Wiley, 2011. https://doi.org/10.1002/9781119998518.

White JC, Tompalski P, Bater CW. *et al.* Enhanced forest inventories in Canada: implementation, status, and research needs. *Can J For Res* 2025;**55**:1–37. https://doi.org/10.1139/cjfr-2024-0255.

White JC, Wulder MA, Hermosilla T. *et al.* A nationwide annual characterization of 25 years of forest disturbance and recovery for Canada using Landsat time series. *Remote Sens Environ* 2017;**194**:303–21. https://doi.org/10.1016/j.rse.2017.03.035.

White JC, Wulder MA, Hobart GW. *et al.* Pixel-based image compositing for large-area dense time series applications and science. *Can J Remote Sens* 2014;**40**:192–212. https://doi.org/10.1080/07038992.2014.945827.

Wulder MA, Campbell C, White JCC. *et al.* National circumstances in the international circumboreal community. *For Chron* 2007;**83**:539–56. https://doi.org/10.5558/tfc83539-4.

Wulder MA, Hermosilla T, Stinson G. *et al.* Satellite-based time series land cover and change information to map forest area consistent with national and international reporting requirements. *Forestry: An International Journal of Forest Research* 2020;**93**:331–43. https://doi.org/10.1093/forestry/cpaa006.

Wulder MA, Kurz WA, Gillis M. National level forest monitoring and modeling in Canada. *Prog Plan* 2004;**61**:365–81. https://doi.org/10.1016/S0305-9006(03)00069-2.

Wulder MA, Roy DP, Radeloff VC. *et al.* Fifty years of Landsat science and impacts. *Remote Sens Environ* 2022;**280**:113195. https://doi.org/10.1016/j.rse.2022.113195.

Wulder MA, White JC, Bater CW. *et al.* Lidar plots — a new large-area data collection option: context, concepts, and case study. *Can J Remote Sens* 2012;**38**:600–18. https://doi.org/10.5589/m12-049.

Wulder MA, White Joanne C, Morgan C. *et al.* Monitoring Canada's forests. Part 1: completion of the EOSD land cover project. *Can J Remote Sens* 2008;**34**:549–62. https://doi.org/10.5589/m08-066.

Forestry: An International Journal of Forest Research, 2026, **99**, 1–17
https://doi.org/10.1093/forestry/cpaf067
Original Article